United States Patent No. 12,000,003

Solar-Powered Backpack with Integrated Battery Charger

ABSTRACT

A backpack incorporates a flexible photovoltaic panel mounted on an exterior-facing surface, electrically connected to an internal battery pack and a charge-management circuit. The circuit regulates current delivered to the battery pack from the photovoltaic panel and exposes a standard output port for charging portable electronic devices carried within the backpack.

BACKGROUND

Portable electronic devices frequently require charging while a user is away from fixed power outlets, such as during outdoor travel or commuting. Carrying a separate portable solar panel and battery pack in addition to a backpack adds bulk and requires manual assembly of charging cables each time charging is desired.

SUMMARY

In one embodiment, a flexible photovoltaic panel is laminated to the exterior rear panel of a backpack and connected through a weatherproof conduit to a charge-management circuit housed within an internal compartment. The circuit includes a maximum power point tracking stage and a lithium-ion battery pack, and exposes a USB output port on an interior wall of the compartment for direct charging of a portable device without external cabling.

CLAIMS

1. A backpack comprising: a flexible photovoltaic panel affixed to an exterior surface; a battery pack housed within an internal compartment; a charge-management circuit electrically coupling the photovoltaic panel to the battery pack; and an output port electrically coupled to the battery pack and accessible from within the internal compartment. 2. The backpack of claim 1, wherein the charge-management circuit includes a maximum power point tracking stage. 3. The backpack of claim 1, wherein the output port conforms to a USB electrical interface standard.